

- Q.27 State the concept of shear strength. (CO7)
 Q.28 Write note on Quick sand condition. (CO5)
 Q.29 List any four difference between compaction and consolidation. (CO6)
 Q.30 Define cohesion and angle of internal friction. (CO8)
 Q.31 A soil has water content 18% $G=2.60$, and void ratio 0.65. Find degree of saturation. (Co2)
 Q.32 Define disturbed and undisturbed samples. State their significance. (CO9)
 Q.33 Write short note on isolated footing and raft footing. (CO11)
 Q.34 Describe plate load test procedure. (CO10)
 Q.35 Define pile. Give its classification. (CO11)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain in detail standard penetration test conducted in the field. Give its application. (CO10)
 Q.37 Explain Auger boring and Wash boring. State their advantages and disadvantages. (CO9)
 Q.38 What are the different types of shallow foundations? Explain with diagram. (CO11)

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M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Residual soils are formed by (CO1)
 a) Transportation by wind
 b) Transportation by water
 c) Weathering of rocks at the same place
 d) Deposition of glaciers
- Q.2 The permeability of soil is the measure of its ability to (CO4)
 a) Retain water b) Expand & contract
 c) Transmit water d) Resist shear
- Q.3 A disturbed soil sample in one in which (CO9)
 a) Natural structure is preserved
 b) Layers are clearly visible
 c) Natural structure is altered
 b) It is tested in situ

- Q.4 The shear strength of cohesive soil is mainly due to (CO7)
- a) Friction b) Cohesion
- c) Particle size d) Permeability
- Q.5 Effective stress is also known as (CO5)
- a) Principal stress b) Pore pressure
- c) Inter granular stress d) None of these
- Q.6 Neutral stress refers to (CO5)
- a) Submerged weight of soil
- b) Pore water pressure
- c) Saturated weight of soil
- d) Minor principal stress
- Q.7 One of the methods of soil stabilization is (CO10)
- a) Excavation b) Compaction
- c) Grouting d) None of these
- Q.8 Uniformity coefficient is the ratio of (CO3)
- a) D_{10}/D_{60} b) D_{60}/D_{10}
- c) D_{30}/D_{10} d) D_{30}/D_{60}
- Q.9 The unit of dry density is (CO2)
- a) Kg/m^3 b) N/m^2
- c) gm/cm^2 d) m/s
- Q.10 Foundation that spreads load over large area is (CO11)
- a) Pile b) Raft
- c) Pier d) Strip

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Clay has _____ cohesion. (CO3)
- Q.12 _____ test is used to find liquid limit. (CO3)
- Q.13 A mat foundation is also known as _____ foundation. (CO11)
- Q.14 Porosity and void ratio are the same. True/False (CO2)
- Q.15 _____ is the percentage of voids in total volume. (CO2)
- Q.16 Core cutter is used in rocky soil. True/False (CO8)
- Q.17 _____ foundation is suitable for expansive soils. (CO11)
- Q.18 Creep occurs more in _____ soil. (CO6)
- Q.19 _____ is used to measure shear strength. (CO7)
- Q.20 Lime stabilization is used for clayey soils. True/False (CO10)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define Loess, Talus and Till. (CO1)
- Q.22 Define shear strength. Give its importance. (CO7)
- Q.23 State BIS soil classification system briefly. (CO3)
- Q.24 State the advantages and disadvantages of direct shear test. (any five) (CO7)
- Q.25 Explain location, depth and spacing of soil exploration. (CO9)
- Q.26 Define permeability. State any three factors affecting it. (CO4)